Latest attempt

Values for j=4 (NEW!)

In[∘]:=
```
(*J11+J31*)
```

In[∘]:=
```
J11 = (1 - Exp[-x^(-a)]) * Integrate[Exp[-y^(-a)] * D[y^(-a * w) Log[y], y],
    {y, 0, x}, Assumptions → a > 0 && 0 < w < 1 && x > 0]
```

Out[∘]=
$$\frac{\left(1-e^{-x^{-a}}\right)\left(\text{Gamma}[w,\ x^{-a}]\left(1-a\,w\,\text{Log}[x]\right)+w\,\text{MeijerG}[\{\{\},\ \{1,\ 1\}\},\ \{\{0,\ 0,\ w\},\ \{\}\},\ x^{-a}]\right)}{a}$$

In[∘]:=
```
J31 = Exp[-x^(-a)] * Integrate[(1 - Exp[-y^(-a)]) * D[y^(-a * w) Log[y], y],
    {y, x, Infinity}, Assumptions → a > 0 && 0 < w < 1 && x > 0]
```

Out[∘]= $Aborted

In[∘]:=
```
Integrate[(J11 + J31) / x // Simplify, {x, 0, Infinity}, Assumptions → a > 0 && 0 < w < 1]
```

Out[∘]=
$$\text{Integrate}\Big[\frac{1}{a\,x}\Big(\text{Gamma}[w,\ x^{-a}]-a\,e^{-x^{-a}}\,x^{-a\,w}\,\text{Log}[x]-a\,w\,\text{Gamma}[w,\ x^{-a}]\,\text{Log}[x]+w\,\text{MeijerG}[\{\{\},\ \{1,\ 1\}\},\ \{\{0,\ 0,\ w\},\ \{\}\},\ x^{-a}]-e^{-x^{-a}}\,\text{Gamma}[w]\left(1+w\,\text{PolyGamma}[0,\ w]\right)\Big),\ \{x,\ 0,\ \infty\},\ \text{Assumptions}\to a>0\ \&\&\ 0<w<1\Big]$$

```
(*J12+J32*)
```

In[∘]:=
```
J12 = (1 - Exp[-x^(-a)]) * Integrate[Exp[-y^(-a)] * D[y^(-a * w), y],
    {y, 0, x}, Assumptions → a > 0 && x > 0 && 0 < w < 1]
```

Out[∘]=
$$-\left(\left(1-e^{-x^{-a}}\right)w\,\text{Gamma}[w,\ x^{-a}]\right)$$

In[∘]:=
```
J32 = Exp[-x^(-a)] * Integrate[(1 - Exp[-y^(-a)]) * D[y^(-a * w), y],
    {y, x, Infinity}, Assumptions → a > 0 && x > 0 && 0 < w < 1]
```

Out[∘]=
$$e^{-x^{-a}}\left(-x^{-a\,w}+w\,\text{Gamma}[w]-w\,\text{Gamma}[w,\ x^{-a}]\right)$$

In[∘]:=
```
Integrate[(J12 + J32) / x // Simplify, {x, 0, Infinity}, Assumptions → a > 0 && 0 < w < 1]
```

Out[∘]=
$$\text{Integrate}\Big[e^{-x^{-a}}\,x^{-1-a\,w}\left(-1+w\,x^{a\,w}\,\text{Gamma}[w]-e^{x^{-a}}\,w\,x^{a\,w}\,\text{Gamma}[w,\ x^{-a}]\right),\ \{x,\ 0,\ \infty\},\ \text{Assumptions}\to a>0\ \&\&\ 0<w<1\Big]$$

```
(*J13+J33*)
```

In[∘]:=
```
J13 = (1 - Exp[-x^(-a)]) *
  Integrate[Exp[-y^(-a)] * D[Log[y], y], {y, 0, x}, Assumptions → a > 0 && x > 0]
```

Out[∘]=
$$\frac{\left(1-e^{-x^{-a}}\right)\text{Gamma}[0,\ x^{-a}]}{a}$$

In[•]:=
```
J33 = Exp[-x^(-a)] *
  Integrate[(1 - Exp[-y^(-a)]) * D[Log[y], y], {y, x, Infinity}, Assumptions → a > 0 && x > 0]
```

Out[•]=
$$\frac{e^{-x^{-a}}\left(\text{EulerGamma} - \text{ExpIntegralEi}[-x^{-a}] - a\,\text{Log}[x]\right)}{a}$$

In[•]:=
```
FullSimplify[(J13 + J33) / x, Assumptions → a > 0 && x > 0]
```

Out[•]=
$$\frac{e^{-x^{-a}}\left(\text{EulerGamma} - e^{x^{-a}}\,\text{ExpIntegralEi}[-x^{-a}] - a\,\text{Log}[x]\right)}{a\,x}$$

In[•]:=
$$\text{Integrate}\left[\frac{e^{-x^{-a}}\left(\text{EulerGamma} - e^{x^{-a}}\,\text{ExpIntegralEi}[-x^{-a}] - a\,\text{Log}[x]\right)}{a\,x},\ \{x, 0, \infty\},\ \text{Assumptions} \to a > 0\right]$$

Out[•]=
$$\frac{\pi^2}{6\,a^2}$$

schmier

In[•]:=
$$\text{Integrate}\left[e^{-x^{-a}}\,x^{-1-a\,w}\,(-1),\ \{x, 0, \infty\},\ \text{Assumptions} \to a > 0\ \&\&\ 0 < w < 1\right]$$

Out[•]=
$$-\frac{\text{Gamma}[w]}{a}$$

In[•]:=
```
Integrate[Exp[-y^(-a)] * y^(-a * w - 1) Log[y],
  {y, 0, x}, Assumptions → a > 0 && 0 < w < 1 && x > 0]
```

Out[•]=
$$\frac{a\,\text{Gamma}[w, x^{-a}]\,\text{Log}[x] - \text{MeijerG}[\{\{\},\{1,1\}\},\{\{0,0,w\},\{\}\},x^{-a}]}{a^2}$$

In[•]:=
```
Integrate[(1 - Exp[-y^(-a)]) * y^(-a * w - 1) Log[y],
  {y, x, Infinity}, Assumptions → a > 0 && 0 < w < 1 && x > 0]
```

Out[•]=
$$\frac{1}{a^2}\left(a\,\text{Gamma}[w, x^{-a}]\,\text{Log}[x] + \frac{x^{-a\,w}\left(1 + a\,w\,\text{Log}[x]\right)}{w^2} - \text{MeijerG}[\{\{\},\{1,1\}\},\{\{0,0,w\},\{\}\},x^{-a}] + \text{Gamma}[w]\,\text{PolyGamma}[0, w]\right)$$

In[•]:=
$$\text{Integrate}\left[\left(\text{Gamma}[w]\,\text{Exp}[-x] - \text{Gamma}[w, x]\right)/x,\ \{x, 0, \infty\},\ \text{Assumptions} \to a > 0\ \&\&\ 0 < w < 1\right]$$

Out[•]=
$$-\text{Gamma}[w]\,(\text{EulerGamma} + \text{PolyGamma}[0, w])$$

In[∘]:=
```
Integrate[((Gamma[w] Exp[-x^(-a)] - Gamma[w, x^(-a)]) / x,
  {x, 0, ∞}, Assumptions → a > 0 && 0 < w < 1]
```

Out[∘]= $\text{Integrate}\left[\frac{e^{-x^{-a}}\,\text{Gamma}[w]-\text{Gamma}[w,\,x^{-a}]}{x},\ \{x,\,0,\,\infty\},\ \text{Assumptions}\to a>0\ \&\&\ 0<w<1\right]$

In[∘]:=
```
D[Gamma[w, u], w]
```

Out[∘]= Gamma[w, u] Log[u] + MeijerG[{{}, {1, 1}}, {{0, 0, w}, {}}, u]

In[∘]:= $\text{Integrate}\left[\frac{e^{-x^{-a}}\,x^{\wedge}(-a\,w)\,(1+\text{Log}[x^{\wedge}(a\,w)])}{x},\ \{x,\,0,\,\infty\},\ \text{Assumptions}\to a>0\ \&\&\ 0<w<1\right]$

Out[∘]= $\frac{\text{Gamma}[w]\,(1-w\,\text{PolyGamma}[0,\,w])}{a}$

In[∘]:= $\text{Integrate}\left[\frac{e^{-x}\,\text{Gamma}[w]\,\text{PolyGamma}[0,\,w]-\text{D}[\text{Gamma}[w,\,x],\,w]}{x},\ \{x,\,0,\,\infty\},\ \text{Assumptions}\to 0<w<1\right]$

Out[∘]= $-\text{Gamma}[w]\,(\text{PolyGamma}[0,\,w]\,(\text{EulerGamma}+\text{PolyGamma}[0,\,w])+\text{PolyGamma}[1,\,w])$

Values for j=4 (OLD!)

```
(*J11+J31*)
```

In[∘]:=
```
J11 = (1 - Exp[-x^(-a)]) *
  Integrate[Exp[-y^(-a)] * D[y^(-a) Log[y], y], {y, 0, x}, Assumptions → a > 0]
```

Out[∘]= $\frac{(1-e^{-x^{-a}})\,(-\text{ExpIntegralEi}[-x^{-a}]+e^{-x^{-a}}\,(1-a\,\text{Log}[x]))}{a}$

In[∘]:=
```
J31 = Exp[-x^(-a)] *
  Integrate[(1 - Exp[-y^(-a)]) * D[y^(-a) Log[y], y], {y, x, Infinity}, Assumptions → a > 0]
```

Out[∘]= $\frac{e^{-x^{-a}}\,(-1+e^{-x^{-a}}+\text{EulerGamma}-\text{ExpIntegralEi}[-x^{-a}]+a\,(-e^{-x^{-a}}-x^{-a})\,\text{Log}[x])}{a}$

In[∘]:=
```
Integrate[(J11 + J31) / x // Simplify, {x, 0, Infinity}, Assumptions → a > 0]
```

Out[∘]= $\frac{-6\,\text{EulerGamma}+\pi^2}{6\,a^2}$

```
(*J12+J32*)
```

In[∘]:=
```
J12 = (1 - Exp[-x^(-a)]) *
  Integrate[Exp[-y^(-a)] * D[y^(-a), y], {y, 0, x}, Assumptions → a > 0 && x > 0]
```

Out[∘]= $-e^{-x^{-a}}\,(1-e^{-x^{-a}})$

In[∘]:=
```
J32 = Exp[-x^(-a)] *
  Integrate[(1 - Exp[-y^(-a)]) * D[y^(-a), y], {y, x, Infinity}, Assumptions → a > 0 && x > 0]
```

Out[∘]= $e^{-x^{-a}}\left(1-e^{-x^{-a}}-x^{-a}\right)$

In[∘]:=
```
Integrate[(J12 + J32) / x // Simplify, {x, 0, Infinity}, Assumptions → a > 0]
```

Out[∘]= $-\frac{1}{a}$

```
(*J13+J33*)
```

In[∘]:=
```
J13 = (1 - Exp[-x^(-a)]) *
  Integrate[Exp[-y^(-a)] * D[Log[y], y], {y, 0, x}, Assumptions → a > 0 && x > 0]
```

Out[∘]= $\frac{\left(1-e^{-x^{-a}}\right)\text{Gamma}\left[0,\ x^{-a}\right]}{a}$

In[∘]:=
```
J33 = Exp[-x^(-a)] *
  Integrate[(1 - Exp[-y^(-a)]) * D[Log[y], y], {y, x, Infinity}, Assumptions → a > 0 && x > 0]
```

Out[∘]= $\frac{e^{-x^{-a}}\left(\text{EulerGamma}-\text{ExpIntegralEi}\left[-x^{-a}\right]-a\,\text{Log}[x]\right)}{a}$

In[∘]:=
```
FullSimplify[(J13 + J33) / x, Assumptions → a > 0 && x > 0]
```

Out[∘]= $\frac{e^{-x^{-a}}\left(\text{EulerGamma}-e^{x^{-a}}\,\text{ExpIntegralEi}\left[-x^{-a}\right]-a\,\text{Log}[x]\right)}{a\,x}$

In[∘]:=
```
Integrate[(e^(-x^(-a)) (EulerGamma - e^(x^(-a)) ExpIntegralEi[-x^(-a)] - a Log[x]))/(a x),
 {x, 0, ∞}, Assumptions → a > 0]
```

Out[∘]= $\frac{\pi^2}{6\,a^2}$

Values \sigma_{i,j} for i,j<4

In[∘]:=
```
cdfY[y_, a_, p0_] :=
 Exp[-y^(-a)] * (1 + p0 * y^(-a))(*Piecewise[{{Exp[-y^(-a)]*(a+p0*y^(-a)),y>0},{0,y≤0}}]*)
```

In[∘]:=
```
pdfY[y_, a_, p0_] := D[cdfY[y, a, p0], y]
```

In[∘]:=
```
pdfY[y, a, p0]
```

Out[∘]= $-a\,e^{-y^{-a}}\,p0\,y^{-1-a}+a\,e^{-y^{-a}}\,y^{-1-a}\left(1+p0\,y^{-a}\right)$

In[∘]:=
```
distY[a_, p0_] := ProbabilityDistribution[
  pdfY[y, a, p0], {y, 0, Infinity}, Assumptions → a > 0 && 0 ≤ p0 ≤ 1]
```

```
In[•]:= s11[a_, p0_, p1_, w_] :=
  Expectation[x^(-2*a*w) Log[x]^2, x ≈ distY[a, p0], Assumptions →
      a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] - Expectation[x^(-a*w) Log[x], x ≈ distY[a, p0],
      Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1]^2 // Simplify

In[•]:= s12[a_, p0_, p1_, w_] := Expectation[x^(-2*a*w) Log[x],
     x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] -
    Expectation[Log[x] x^(-a*w), x ≈ distY[a, p0],
      Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] × Expectation[x^(-a*w),
      x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] // Simplify

In[•]:= s13[a_, p0_, p1_, w_] := Expectation[x^(-a*w) Log[x]^2,
     x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] -
    Expectation[ Log[x], x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] ×
     Expectation[x^(-a*w) Log[x], x ≈ distY[a, p0],
      Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] // Simplify

In[•]:= s22[a_, p0_, p1_, w_] := Expectation[x^(-2*a*w), x ≈ distY[a, p0],
     Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] - Expectation[x^(-a*w),
      x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1]^2 // Simplify

In[•]:= s23[a_, p0_, p1_, w_] := Expectation[x^(-a*w) Log[x],
     x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] -
    Expectation[ Log[x], x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] ×
     Expectation[x^(-a*w), x ≈ distY[a, p0],
      Assumptions → a > 0 && 0 ≤ p0 ≤ 1 && 0 ≤ w ≤ 1] // Simplify

In[•]:= s33[a_, p0_, p1_, w_] :=
  Expectation[Log[x]^2, x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1] -
    Expectation[Log[x], x ≈ distY[a, p0], Assumptions → a > 0 && 0 ≤ p0 ≤ 1]^2 // Simplify
```

sanity check: schmier

In[•]:=
```
s14[a, p0, p1, w] -
  Gamma[w + 1] / a ^ 2 ((p0 EulerGamma + p0 PolyGamma[0, w + 1] + p1) (1 + w PolyGamma[0, w + 1]) +
      p0 w PolyGamma[1, w + 1]) -
  Gamma[w] / a ^ 2 (EulerGamma + 2 PolyGamma[0, w] + w PolyGamma[0, w] ^ 2 +
      w EulerGamma PolyGamma[0, w] + w PolyGamma[1, w]) // FullSimplify
```

Out[•]=
$$-\frac{1}{a^2}\text{Gamma}[w]\left(\text{EulerGamma}+\text{PolyGamma}[0,w]\left(2+w\left(\text{EulerGamma}+\text{PolyGamma}[0,w]\right)\right)+w\,\text{PolyGamma}[1,w]\right)-\frac{\text{Gamma}[1+w]\left(\left(p1+p0\,\text{HarmonicNumber}[w]\right)\left(1+w\,\text{PolyGamma}[0,1+w]\right)+p0\,w\,\text{PolyGamma}[1,1+w]\right)}{a^2}+\text{s14}[a,p0,p1,w]$$

In[•]:=
```
Gamma[1] + 1 Gamma[1] (EulerGamma + PolyGamma[0, 1])
```

Out[•]= 1

In[•]:=
```
s13[1, 0, 0, 1]
```

Out[•]= $\frac{1}{6}\left(-6\,\text{EulerGamma}+\pi^2\right)$

Assembling the Covariance Matrix

In[•]:=
```
(*outdated: Sigma[a_,p0_,p1_]:=
 {{1/(3 a^2) (3-12 p0^2+EulerGamma^2 (3+6 p0-3 p0^2)+6 EulerGamma (-2-5 p0+2 p0^2)+π^2+
      2 p0 (9+π^2)), 1/a (-2+EulerGamma-5 p0+2 EulerGamma p0+2 p0^2-EulerGamma p0^2),
   1/(6 a^2) (-12 p0^2+6 EulerGamma (-1-p0+p0^2)+π^2+p0 (6+π^2)),
   (π^2-6 EulerGamma)/(6 a^2)+(π^2/6+1-EulerGamma)p0/a^2+(2-EulerGamma) p1/a^2},
  {1/a (-2+EulerGamma-5 p0+2 EulerGamma p0+2 p0^2-EulerGamma p0^2),
   1+2 p0-p0^2, (-1-p0+p0^2)/a, -1/a-(p0+p1)/a},
  {1/(6 a^2) (-12 p0^2+6 EulerGamma (-1-p0+p0^2)+π^2+p0 (6+π^2)),
   (-1-p0+p0^2)/a, (-6 p0^2+π^2)/(6 a^2), (π^2+6p1)/(6 a^2)},
  {(π^2-6 EulerGamma)/(6 a^2)+(π^2/6+1-EulerGamma)p0/a^2+(2-EulerGamma) p1/a^2,
   -1/a-(p0+p1)/a, (π^2+6p1)/(6 a^2), π^2/(6 a^2)}}*)
```

In[∘]:=
```
s14[a_, p0_, p1_, w_] :=
 Gamma[w + 1] / a ^ 2 ((p0 EulerGamma + p0 PolyGamma[0, w + 1] + p1) (1 + w PolyGamma[0, w + 1]) +
     p0 w PolyGamma[1, w + 1])(*w/a^2
   (p1*Gamma[w+1](1/w+PolyGamma[0,w+1])+p0 Gamma[w](2PolyGamma[0,w+1]+w PolyGamma[0,w]
         (PolyGamma[0,w+1]+EulerGamma)+w PolyGamma[1,w+1]+2 EulerGamma))*) +
  Gamma[w] / a ^ 2 (EulerGamma + 2 PolyGamma[0, w] + w PolyGamma[0, w] ^ 2 +
     w EulerGamma PolyGamma[0, w] + w PolyGamma[1, w])
    (*Gamma[w](p0+w p1)/a^2-p0 w /a^2(Gamma[w+2]PolyGamma[0,w+2]/w-
       Gamma[w]PolyGamma[0,w]-p1/p0 Gamma[w+1]PolyGamma[0,w+1])*)
```

In[∘]:=
```
s14[a, p0, p1, w] // FullSimplify
```

Out[∘]=
$$\frac{1}{a^2}\,\text{Gamma}[w]\,\Big(\text{EulerGamma} + p0 + 2\,(\text{EulerGamma}\,p0 + p1)\,w + \big(2 + w\,(\text{EulerGamma} + 3\,p0 + \text{EulerGamma}\,p0\,w + p1\,w)\big)\,\text{PolyGamma}[0,\,w] + w\,(1 + p0\,w)\,\text{PolyGamma}[0,\,w]^2 + w\,(1 + p0\,w)\,\text{PolyGamma}[1,\,w]\Big)$$

In[∘]:=
```
s24[a_, p0_, p1_, w_] := -w ^ 2 / a (EulerGamma + PolyGamma[0, w + 1]) Gamma[w] p0 -
  w / a Gamma[w + 1] p1 - Gamma[w] / a - w / a Gamma[w] (EulerGamma + PolyGamma[0, w])
   (*p0*Gamma[w+2]-Gamma[w+1](p0+w p1)-Gamma[w]-w Gamma[w](EulerGamma+PolyGamma[0,w])*)
```

In[∘]:=
```
s34[a_, p0_, p1_, w_] := Pi ^ 2 / (6 a ^ 2) + p1 / a ^ 2
```

In[∘]:=
```
Sigma[a_, p0_, p1_, w_] :=
 {{s11[a, p0, p1, w], s12[a, p0, p1, w], s13[a, p0, p1, w], s14[a, p0, p1, w]},
  {s12[a, p0, p1, w], s22[a, p0, p1, w], s23[a, p0, p1, w], s24[a, p0, p1, w]},
  {s13[a, p0, p1, w], s23[a, p0, p1, w], s33[a, p0, p1, w], s34[a, p0, p1, w]},
  {s14[a, p0, p1, w], s24[a, p0, p1, w], s34[a, p0, p1, w], π^2 / (6 a ^ 2)}}
```

In[∘]:=
```
Sigma[a, p0, p1, w] // MatrixForm
```

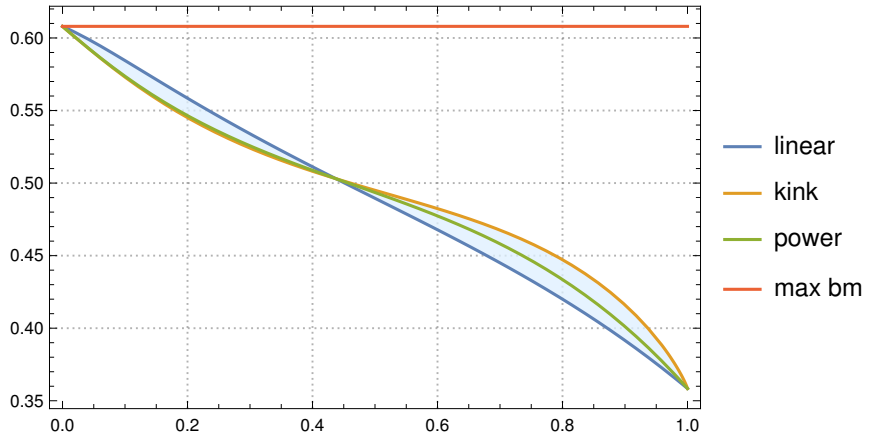
Out[∘]//MatrixForm=
$$\begin{pmatrix} \frac{-\text{Gamma}[1+w]^2\,(p0+(1+p0\,w)\,\text{PolyGamma}[0,1+w])^2+\text{Gamma}[1+2\,w]\,(2\,p0\,\text{PolyGamma}[0,1+2\,w]+(1+2\,p0\,w)\,\text{PolyGamm}}{a^2} \\ \frac{(1+p0\,w)\,\text{Gamma}[1+w]^2\,(p0+(1+p0\,w)\,\text{PolyGamma}[0,1+w])-\text{Gamma}[1+2\,w]\,(p0+(1+2\,p0\,w)\,\text{P}}{a} \\ \frac{(\text{EulerGamma}-p0)\,\text{Gamma}[1+w]\,(p0+(1+p0\,w)\,\text{PolyGamma}[0,1+w])+w\,\text{Gamma}[w]\,(2\,p0\,\text{PolyGamma}[0,1+w]+(1+p0\,w)\,\text{Po}}{a^2} \\ \frac{\text{Gamma}[w]\,(\text{EulerGamma}+2\,\text{PolyGamma}[0,w]+\text{EulerGammaw}\,\text{PolyGamma}[0,w]+w\,\text{PolyGamma}[0,w]^2+w\,\text{PolyGamma}[1,w])}{a^2} + \frac{\text{Gamma}[1+w]\,((\text{EulerGamma}}{} \end{pmatrix}$$

In[∘]:=
```
Sigma[a_, p0_, p1_, w_] :=
```
$$\begin{pmatrix} \frac{-\text{Gamma}[1+w]^2\,(p0+(1+p0\,w)\,\text{PolyGam}}{} \\ \frac{(1+p0\,w)\,\text{Ga}}{} \\ \frac{(\text{EulerGamma}-p0)\,\text{Gamma}[1+w]\,(p0+(1+}{} \\ \frac{\text{Gamma}[w]\,(\text{EulerGamma}+2\,\text{PolyGamma}[0,w]+\text{EulerGammaw}\,\text{PolyGamma}[0,w]+w\,\text{PolyGamma}[0,w]^2+w\,\text{Poly}}{a^2} \end{pmatrix}$$

```
In[•]:= (*define upsilon und Psi dot first*)
        Y[x_, p0_] := p0 * Gamma[x + 2] + (1 - p0) Gamma[x + 1]

In[•]:= PI[x_, p0_] :=
          1 / x - ((1 - p0) Gamma[1 + x] PolyGamma[0, 1 + x] + p0 Gamma[2 + x] PolyGamma[0, 2 + x]) / Y[x, p0] +
           p0 / 2 - EulerGamma

In[•]:= ΨDot[a_, p0_, w_] := -2 / (w * a) ^ 2 -
           2 / (a ^ 2 * Y[w, p0] ^ 2) * ((D[D[Y[z, p0], z], z] /. z → w) * Y[w, p0] - (D[Y[z, p0], z] /. z → w) ^ 2)

In[•]:= z0[p0_, w_] := Y[w, p0] / 2

In[•]:= s1[a_, p0_, w_] := z0[p0, w] ^ (-1 / (a * w))

In[•]:= M[a_, p0_, w_] :=
          1 / ΨDot[a, p0, w] * {{-2 / Y[w, p0], -2 (D[Y[z, p0], z] /. z → w) / (a * Y[w, p0] ^ 2), 1, 1},
            z0[p0, w] ^ (-1 / (a * w)) / (a * w) *
             {(D[Y[z, p0], z] /. z → w) / (2 * a * z0[p0, w] ^ 2) - Log[z0[p0, w]] / (a * w * z0[p0, w]),
              (D[Y[z, p0], z] /. z → w) ^ 2 / (4 * a ^ 2 * z0[p0, w] ^ 3) - (D[Y[z, p0], z] /. z → w)
                 Log[z0[p0, w]] / (2 * a ^ 2 * w * z0[p0, w] ^ 2) - ΨDot[a, p0, w] / (2 z0[p0, w]),
              Log[z0[p0, w]] / (a * w) - (D[Y[z, p0], z] /. z → w) / (2 * a * z0[p0, w]),
              Log[z0[p0, w]] / (a * w) - (D[Y[z, p0], z] /. z → w) / (2 * a * z0[p0, w])}}

In[•]:= Mbc[a_, p0_, w_] := {{1 / w, 0}, {-Log[z0[p0, w]] / (a ^ 2 w ^ 2), 1 / s1[a, p0, w]}}

        (*Mbc[a_,p0_,w_]:=
         {{1/w,0},{-Log[(2/Y[w,p0])^(1/(w*a))]/(a*w*(2/Y[w,p0])^(1/(w*a))),1/(2/Y[w,p0])^(1/(w*a))}}*)
        (*below is same but shorter*)

In[•]:= Mbc[a_, p0_, w_] := {{1 / w, 0}, {Log[s1[a, p0, w]] / (a w), 1 / s1[a, p0, w]}}

        (*Mbc[a_,p0_,w_]:={{1,0},{0,1}}(*Mbc[a_,p0_,w_]:={{1,0},{0,1}}(*without biascorr*)*)*)

In[•]:= Mbc[1, 0, 1].{{1, 0}, {-Log[4], 2}} // FullSimplify
Out[•]= {{1, 0}, {0, 1}}

In[•]:= FinalSigma[a_, p0_, p1_, w_] := Mbc[a, p0, w].M[a, p0, w].
           Sigma[a, p0, p1, w].Transpose[M[a, p0, w]].Transpose[Mbc[a, p0, w]]

In[•]:= wp[p_?NumericQ] := Module[{y}, y /. FindRoot[PI[y, p] == 0, {y, 1.0}]];

In[•]:= testp0 = 0
Out[•]= 0

In[•]:= FinalSigma[1, testp0, -testp0, wp[testp0]]
Out[•]= {{0.607927, -1.35681}, {-1.35681, 9.21069}}
```

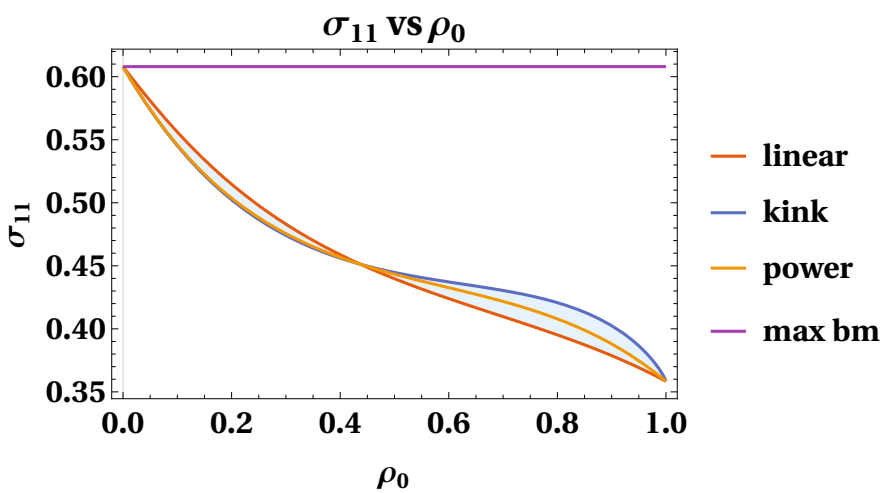
In[•]:=
```
Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 1]]],
  Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]]],
  Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[1, 1]]],
  0.608},
 {p0, 0, 1}, PlotLegends → {"linear", "kink", "power", "max bm"},
 AxesLabel → {"p0", "(1,1) entry"}, PlotRange → All, PlotTheme → "Detailed",
 Filling → {1 → {2}}, FillingStyle → Directive[Opacity[0.75], LightBlue]]
```

Out[•]=



In[•]:=
```
Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 1]]],
  Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]]],
  Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[1, 1]]], 0.608}, {p0, 0, 1},
 PlotTheme → "Scientific", PlotLegends → {"linear", "kink", "power", "max bm"},
 AxesLabel → {"p0", "(1,1) entry"},
 PlotLabel → TraditionalForm@Row[{Subscript[σ, 11], " vs ", Subscript[ρ, 0]}],
 LabelStyle → {16, Bold, FontFamily → "Times"}, PlotRange → All, Filling → {1 → {2}},
 FillingStyle → Directive[Opacity[0.75], RGBColor[0.87, 0.94, 1]], FrameLabel →
  {TraditionalForm@Subscript[ρ, 0], TraditionalForm@Subscript[σ, 11]}]
```

Out[•]=



In[•]:=
```
Export["/home/erik/Downloads/var_alpha_corrected_nobc.pdf",
 Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 1]]],
   Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]]],
   Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[1, 1]]], 0.608}, {p0, 0, 1},
  PlotTheme → "Scientific", PlotLegends → {"linear", "kink", "power", "max bm"},
  AxesLabel → {"p0", "(1,1) entry"},
  PlotLabel → TraditionalForm@Row[{Subscript[σ, 11], " vs ", Subscript[ρ, 0]}],
  LabelStyle → {16, Bold, FontFamily → "Times"}, PlotRange → All, Filling → {1 → {2}},
  FillingStyle → Directive[Opacity[0.75], RGBColor[0.87, 0.94, 1]], FrameLabel →
   {TraditionalForm@Subscript[ρ, 0], TraditionalForm@Subscript[σ, 11]}], "PDF"]
```

Out[•]= /home/erik/Downloads/var_alpha_corrected_nobc.pdf

In[•]:=
```
Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[2, 1]]],
   Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 1]]],
   Evaluate[FinalSigma[1, p0, -p0 ^ 2, wp[p0]][[2, 1]]],
   -0.257},
  {p0, 0, 1}, PlotTheme → "Scientific",
  PlotLegends → {"linear", "kink", "power", "max bm"},
  AxesLabel → {"p0", "(1,1) entry"},
  PlotLabel → TraditionalForm@Row[{Subscript[σ, 12], " vs ", Subscript[ρ, 0]}],
  LabelStyle → {16, Bold, FontFamily → "Times"}, PlotRange → All, Filling → {1 → {2}},
  FillingStyle → Directive[Opacity[0.75`], RGBColor[0.87`, 0.94`, 1]], FrameLabel →
   {TraditionalForm@Subscript[ρ, 0], TraditionalForm@Subscript[σ, 12]}]
```

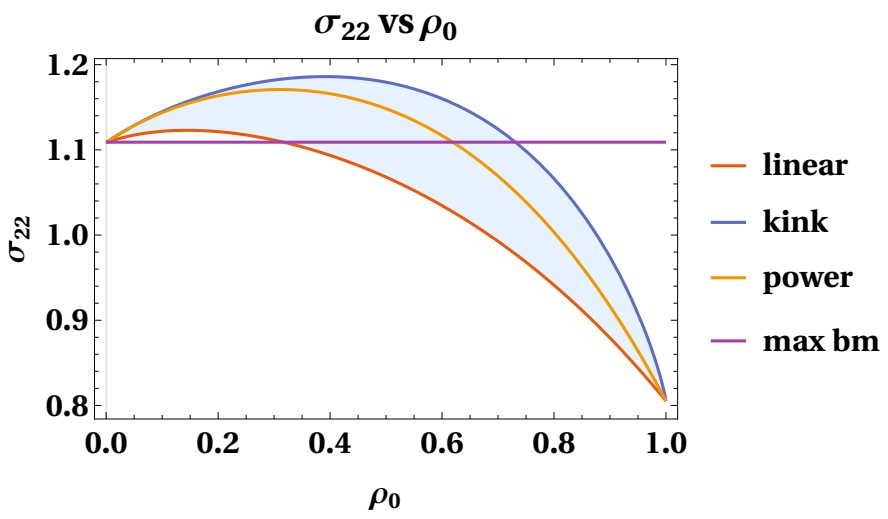
Out[•]= $Aborted

In[•]:=
```
Export["/home/erik/Downloads/cov_alphasigma_corrected_nobc.pdf",
 Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[2, 1]]],
   Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 1]]],
   Evaluate[FinalSigma[1, p0, -p0 ^ 2, wp[p0]][[2, 1]]],
   -0.257},
  {p0, 0, 1}, PlotTheme → "Scientific",
  PlotLegends → {"linear", "kink", "power", "max bm"},
  AxesLabel → {"p0", "(1,1) entry"},
  PlotLabel → TraditionalForm@Row[{Subscript[σ, 12], " vs ", Subscript[ρ, 0]}],
  LabelStyle → {16, Bold, FontFamily → "Times"}, PlotRange → All, Filling → {1 → {2}},
  FillingStyle → Directive[Opacity[0.75], RGBColor[0.87, 0.94, 1]], FrameLabel →
   {TraditionalForm@Subscript[ρ, 0], TraditionalForm@Subscript[σ, 12]}], "PDF"]
```

Out[•]= /home/erik/Downloads/cov_alphasigma_corrected_nobc.pdf

In[•]:=
```
Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[2, 2]]],
  Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 2]]],
  Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[2, 2]]], 1.109},
 {p0, 0, 1}, PlotTheme → "Scientific",
 PlotLegends → {"linear", "kink", "power", "max bm"},
 AxesLabel → {"p0", "(1,1) entry"},
 PlotLabel → TraditionalForm@Row[{Subscript[σ, 22], " vs ", Subscript[ρ, 0]}],
 LabelStyle → {16, Bold, FontFamily → "Times"}, PlotRange → All, Filling → {1 → {2}},
 FillingStyle → Directive[Opacity[0.75`], RGBColor[0.87`, 0.94`, 1]], FrameLabel →
  {TraditionalForm@Subscript[ρ, 0], TraditionalForm@Subscript[σ, 22]}]
```

Out[•]=



In[•]:=
```
Export["/home/erik/Downloads/var_sigma_corrected_nobc.pdf",
 Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[2, 2]]],
   Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 2]]],
   Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[2, 2]]], 1.109},
  {p0, 0, 1}, PlotTheme → "Scientific",
  PlotLegends → {"linear", "kink", "power", "max bm"},
  AxesLabel → {"p0", "(1,1) entry"},
  PlotLabel → TraditionalForm@Row[{Subscript[σ, 22], " vs ", Subscript[ρ, 0]}],
  LabelStyle → {16, Bold, FontFamily → "Times"}, PlotRange → All, Filling → {1 → {2}},
  FillingStyle → Directive[Opacity[0.75], RGBColor[0.87, 0.94, 1]], FrameLabel →
   {TraditionalForm@Subscript[ρ, 0], TraditionalForm@Subscript[σ, 22]}], "PDF"]
```

Out[•]= /home/erik/Downloads/var_sigma_corrected_nobc.pdf

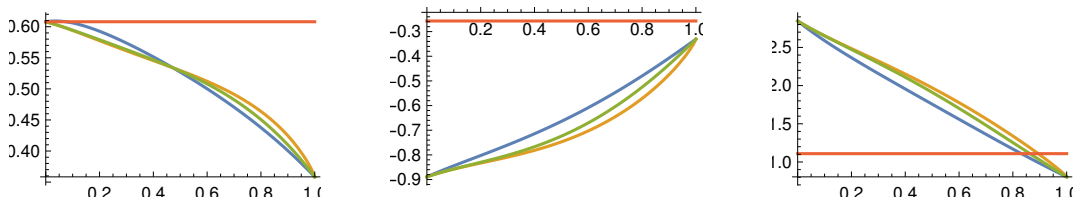
In[•]:=
```
plot11 := Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 1]]],
   Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]]],
   Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[1, 1]]], 0.608}, {p0, 0, 1}]
```

In[∘]:=
```
plot12 := Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 2]]],
    Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 2]]],
    Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[1, 2]]], -0.257}, {p0, 0, 1}]
```

In[∘]:=
```
plot22 := Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[2, 2]]],
    Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 2]]],
    Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[2, 2]]], 1.109}, {p0, 0, 1}]
```

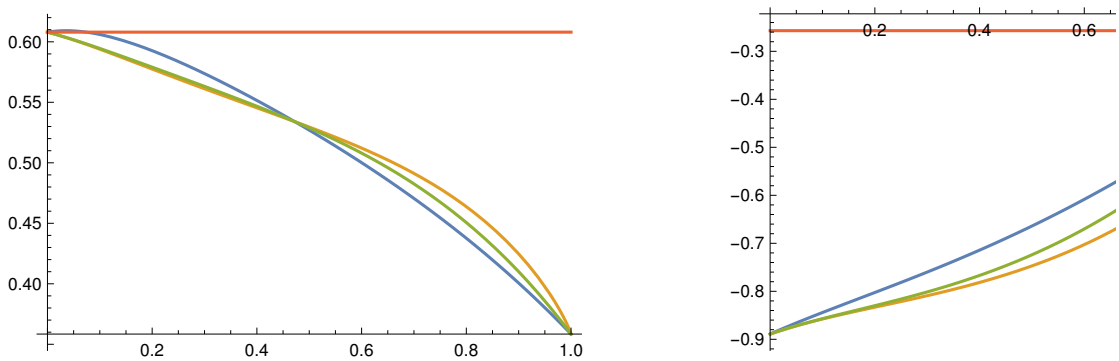
In[∘]:=
```
row = GraphicsRow[{plot11, plot12, plot22}, Spacings → Scaled[.3]]
(*Legended[row,Placed[LineLegend[{"linear","kink","power","max bm"}],Below]]*)
```

Out[∘]=



In[∘]:=
```
Show[row, ImageSize → {1138, 210}, AspectRatio → Full,
 PlotTheme → "Scientific", FrameLabel → {"p0", "(1,1) entry"}]
```

Out[∘]=

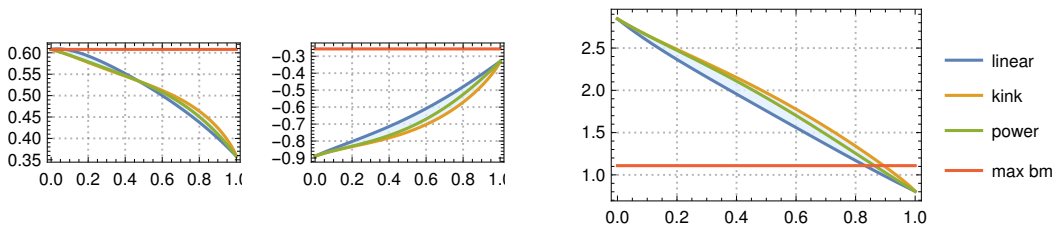


In[∘]:=
```
Export["/home/erik/Downloads/var.pdf", %62, "PDF"]
```

Out[∘]= /home/erik/Downloads/var.pdf

In[∘]:=
```
GraphicsRow[{Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 1]],
    Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]],
    Evaluate[FinalSigma[1, p0, -p0 ^ 2, wp[p0]][[1, 1]], 0.608},
   {p0, 0, 1}, PlotLegends → None, AxesLabel → {"p0", "(1,2) entry"},
   PlotRange → All, PlotTheme → "Detailed", Filling → {1 → {2}},
   FillingStyle → Directive[Opacity[0.75], LightBlue]],
  Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 2]],
    Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 2]],
    Evaluate[FinalSigma[1, p0, -p0 ^ 2, wp[p0]][[1, 2]], -0.257},
   {p0, 0, 1}, AxesLabel → {"p0", "(1,2) entry"}, PlotRange → All,
   PlotTheme → "Detailed", Filling → {1 → {2}}, PlotLegends → None,
   FillingStyle → Directive[Opacity[0.75], LightBlue]],
  Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[2, 2]],
    Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 2]],
    Evaluate[FinalSigma[1, p0, -p0 ^ 2, wp[p0]][[2, 2]], 1.109}, {p0, 0, 1},
   AxesLabel → {"p0", "(2,2) entry"}, PlotRange → All, PlotTheme → "Detailed",
   PlotLegends → {"linear", "kink", "power", "max bm"},
   AxesLabel → {"p0", "(1,1) entry"}, Filling → {1 → {2}},
   FillingStyle → Directive[Opacity[0.75], LightBlue]]}]
```

Out[∘]=



In[∘]:=
```
Integrate[(p (1 - x ^ (1 / p)) - p) / x, {x, 0, 1}]
```

Out[∘]= $-p^2$ if Re[p] > 0

In[∘]:=
```
testp0 = 0.5
```

Out[∘]= 0.5

In[∘]:=
```
testp1 = -testp0 - (1 - testp0) Log[1 - testp0]
```

Out[∘]= -0.153426

In[∘]:=
```
N[Sigma[1, testp0, testp1, wp[testp0]] // MatrixForm]
```

Out[∘]//MatrixForm=

$$\begin{pmatrix} 4.71042 & -2.51753 & 1.64153 & 1.77091 \\ -2.51753 & 1.46347 & -1.1617 & -1.25132 \\ 1.64153 & -1.1617 & 1.39493 & 1.49151 \\ 1.77091 & -1.25132 & 1.49151 & 1.64493 \end{pmatrix}$$

In[∘]:=
```
N[Sigma[1, 0, 0, wp[0]] // MatrixForm]
```

Out[∘]//MatrixForm=

$$\begin{pmatrix} 2.31418 & -1.42278 & 1.06772 & 0. \\ -1.42278 & 1. & -1. & 0. \\ 1.06772 & -1. & 1.64493 & 1.64493 \\ 0. & 0. & 1.64493 & 1.64493 \end{pmatrix}$$

In[∘]:=
```
N[Sigma[1, 1, -1, wp[1]] // MatrixForm]
```

Out[∘]//MatrixForm=

$$\begin{pmatrix} 7.7638 & -3.84557 & 1.71265 & 1.71265 \\ -3.84557 & 2. & -1. & -1. \\ 1.71265 & -1. & 0.644934 & 0.644934 \\ 1.71265 & -1. & 0.644934 & 1.64493 \end{pmatrix}$$

Sanity checks

In[∘]:=
```
datalin = Table[{p0, FinalSigma[1, p0, -p0, wp[p0]][[1, 1]]}, {p0, 0, 1, 0.05}]
```

Out[∘]=
```
{{0., 0.607927}, {0.05, 0.580894}, {0.1, 0.556102}, {0.15, 0.534046},
 {0.2, 0.514686}, {0.25, 0.497785}, {0.3, 0.483042}, {0.35, 0.470151},
 {0.4, 0.45882}, {0.45, 0.448779}, {0.5, 0.439778}, {0.55, 0.431584},
 {0.6, 0.423976}, {0.65, 0.416742}, {0.7, 0.409673}, {0.75, 0.402559},
 {0.8, 0.395187}, {0.85, 0.387333}, {0.9, 0.378761}, {0.95, 0.369221}, {1., 0.35844}}
```

In[∘]:=
```
datakink =
 Table[{p0, FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]]}, {p0, 0, 1, 0.05}]
```

Out[∘]=
```
{{0., 0.607927}, {0.05, 0.573813}, {0.1, 0.545239}, {0.15, 0.521653}, {0.2, 0.502366},
 {0.25, 0.486717}, {0.3, 0.474117}, {0.35, 0.46405}, {0.4, 0.456063},
 {0.45, 0.449752}, {0.5, 0.444741}, {0.55, 0.440674}, {0.6, 0.437191},
 {0.65, 0.433915}, {0.7, 0.430428}, {0.75, 0.426246}, {0.8, 0.420772},
 {0.85, 0.413222}, {0.9, 0.402459}, {0.95, 0.386504}, {1., Indeterminate}}
```

In[∘]:=
```
datapower = Table[{p0, FinalSigma[1, p0, -p0^2, wp[p0]][[1, 1]]}, {p0, 0, 1, 0.05}]
```

Out[∘]=
```
{{0., 0.607927}, {0.05, 0.573991}, {0.1, 0.545792}, {0.15, 0.522608},
 {0.2, 0.503644}, {0.25, 0.488167}, {0.3, 0.475535}, {0.35, 0.465194},
 {0.4, 0.456661}, {0.45, 0.449511}, {0.5, 0.443358}, {0.55, 0.437845},
 {0.6, 0.43263}, {0.65, 0.427375}, {0.7, 0.42174}, {0.75, 0.415374},
 {0.8, 0.407905}, {0.85, 0.398933}, {0.9, 0.388024}, {0.95, 0.374702}, {1., 0.35844}}
```

In[•]:=
```
datalinsig = Table[{p0, FinalSigma[1, p0, -p0, wp[p0]][[2, 2]]}, {p0, 0, 1, 0.05}]
```

Out[•]=
```
{{0., 1.10866}, {0.05, 1.11732}, {0.1, 1.12176}, {0.15, 1.12292},
 {0.2, 1.12137}, {0.25, 1.11747}, {0.3, 1.11146}, {0.35, 1.10346},
 {0.4, 1.09355}, {0.45, 1.08174}, {0.5, 1.06801}, {0.55, 1.05232}, {0.6, 1.0346},
 {0.65, 1.01474}, {0.7, 0.992656}, {0.75, 0.968208}, {0.8, 0.941261},
 {0.85, 0.91166}, {0.9, 0.879241}, {0.95, 0.843825}, {1., 0.805223}}
```

In[•]:=
```
datakinksig =
 Table[{p0, FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[2, 2]]}, {p0, 0, 1, 0.05}]
```

Out[•]=
```
{{0., 1.10866}, {0.05, 1.12873}, {0.1, 1.14491}, {0.15, 1.15797},
 {0.2, 1.16835}, {0.25, 1.17629}, {0.3, 1.18189}, {0.35, 1.18513},
 {0.4, 1.18592}, {0.45, 1.18409}, {0.5, 1.1794}, {0.55, 1.17153}, {0.6, 1.16009},
 {0.65, 1.14457}, {0.7, 1.12435}, {0.75, 1.09859}, {0.8, 1.06622},
 {0.85, 1.02565}, {0.9, 0.974415}, {0.95, 0.907714}, {1., Indeterminate}}
```

In[•]:=
```
datapowersig = Table[{p0, FinalSigma[1, p0, -p0^2, wp[p0]][[2, 2]]}, {p0, 0, 1, 0.05}]
```

Out[•]=
```
{{0., 1.10866}, {0.05, 1.12845}, {0.1, 1.14373}, {0.15, 1.15527},
 {0.2, 1.16348}, {0.25, 1.16859}, {0.3, 1.1707}, {0.35, 1.16982}, {0.4, 1.16588},
 {0.45, 1.15878}, {0.5, 1.14836}, {0.55, 1.13443}, {0.6, 1.11677},
 {0.65, 1.09513}, {0.7, 1.06922}, {0.75, 1.03875}, {0.8, 1.00337},
 {0.85, 0.962732}, {0.9, 0.916441}, {0.95, 0.864085}, {1., 0.805223}}
```

```
Plot[{Evaluate[FinalSigma[1, p0, -p0, wp[p0]][[1, 1]]],
  Evaluate[FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], wp[p0]][[1, 1]]],
  Evaluate[FinalSigma[1, p0, -p0^2, wp[p0]][[1, 1]]], 0.608}, {p0, 0, 1}]
```

In[•]:=
```
N[Pi^2/3 - 1 - EulerGamma]
```

Out[•]= 1.71265

In[•]:=
$$\frac{3 - 12\,p0^2 + \text{EulerGamma}^2\left(3 + 6\,p0 - 3\,p0^2\right) + 6\,\text{EulerGamma}\left(-2 - 5\,p0 + 2\,p0^2\right) + \pi^2 + 2\,p0\left(9 + \pi^2\right)}{3\,a^2}\ /.$$

```
{a → 1, p0 → 1}
```

Out[•]=
$$\frac{1}{3}\left(-9 - 30\,\text{EulerGamma} + 6\,\text{EulerGamma}^2 + \pi^2 + 2\left(9 + \pi^2\right)\right)$$

In[•]:=
```
N[2 EulerGamma - 5]
```

Out[•]= -3.84557

In[•]:=
```
N[3 - 10 EulerGamma + 2 EulerGamma^2 + Pi^2]
```

Out[•]= 7.7638

The bias of the special case

In[•]:=
```
Bf1[w_, p0_] :=
 1/2 Gamma[1 + w] (1 - 2 (-6 + (1 - p0)) (1 - p0) + 4 w + 2 (5 - 2 (1 - p0)) (1 - p0) w - 3 (-1 + (1 - p0)) w^2 +
    (-2 (1 - p0)^2 w (1 + w) + w (1 + w)^2 + (1 - p0) (8 + w (12 - (-5 + w) w))) PolyGamma[0, 1 + w])
```

In[•]:=
```
Bf2[w_, p0_] := 1/2 w ((1 - p0) ((-1 + w) w + 2 (1 - p0) (1 + w)) Gamma[1 + w] - (1 + w) Gamma[2 + w])
```

In[•]:=
```
Bf3[w_, p0_] := 1 / 2 - (1 - p0) ^ 2
```

In[•]:=
```
Bf4[w_, p0_] := 1 / 2 + (1 - p0)
```

In[•]:=
```
B[n_, r_, w_, p0_] :=
 (1 / r) * Mbc[1, p0, w].M[1, p0, w].{Bf1[w, p0], Bf2[w, p0], Bf3[w, p0], Bf4[w, p0]}
```

In[•]:=
```
V[n_, r_, w_, p0_] := (r / n) * FinalSigma[1, p0, -p0, w][[1, 1]]
```

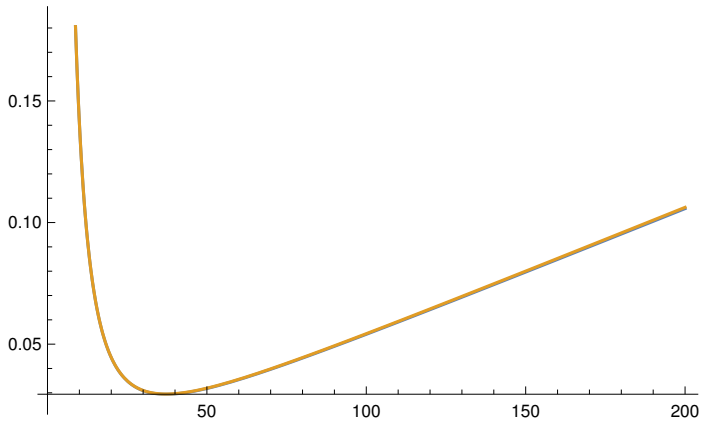
In[•]:=
```
Vmax[n_, r_, w_, p0_] := (r / n) * FinalSigma[1, p0, -p0 - (1 - p0) Log[1 - p0], w][[1, 1]]
```

In[•]:=
```
V[1000, 100, wp[.5], .5] + B[1000, 100, wp[.5], .5][[1]] ^ 2
```

Out[•]=
```
0.0540555
```

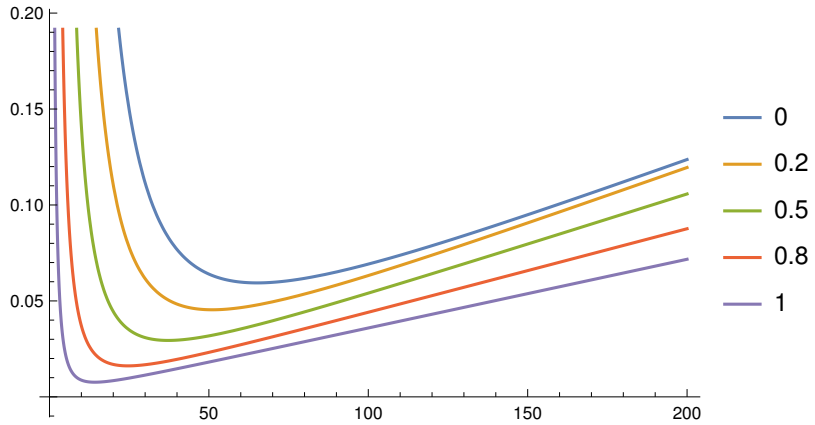
In[•]:=
```
Plot[{V[1000, r, wp[.5], .5] + B[1000, r, wp[.5], .5][[1]] ^ 2,
  Vmax[1000, r, wp[.5], .5] + B[1000, r, wp[.5], .5][[1]] ^ 2}, {r, 1, 200}]
```

Out[•]=



In[•]:=
```
Plot[{V[1000, r, wp[.0], .0] + B[1000, r, wp[.0], .0][[1]]^2,
   V[1000, r, wp[.2], .2] + B[1000, r, wp[.2], .2][[1]]^2,
   V[1000, r, wp[.5], .5] + B[1000, r, wp[.5], .5][[1]]^2,
   V[1000, r, wp[.8], .8] + B[1000, r, wp[.8], .8][[1]]^2,
   V[1000, r, wp[1], 1] + B[1000, r, wp[1], 1][[1]]^2},
  {r, 1, 200}, PlotLegends → {0, .2, .5, .8, 1}]
```

Out[•]=



New Attempt

In[•]:=
```
G[x_, a_] := Exp[-x^(-a)]
```

In[•]:=
```
Gy[x_, a_, ρ_] := Exp[-x^(-a)] (1 + ρ[0] * x^(-a))
```

In[•]:=
```
H[x_, y_, a_, ρ_] := G[y, a] * (1 + ρ[(y / x)^a] * y^(-a))
```

In[•]:=
```
H[x, y, a, ρ]
```

Out[•]= $e^{-y^{-a}}\left(1+y^{-a}\,\rho\left[\left(\frac{y}{x}\right)^{a}\right]\right)$

In[•]:=
```
ρiid[x_] := Piecewise[{{1 - x, 0 < x < 1}}]
```

In[•]:=
```
ρiid[x_] := (1 - x^2) / 2
```

In[•]:=
```
f[y_, a_] := Log[y]
```

In[•]:=
```
Integrate[(H[x, y, a, ρiid] - G[x, a] × Gy[y, a, ρiid]) * (D[f[y, a], y] / x),
   {y, 0, x}, Assumptions → a > 0 && x > 0] + G[x, a] * Integrate[
    (1 - Gy[y, a, ρiid]) * (D[f[y, a], y] / x), {y, x, Infinity}, Assumptions → a > 0 && x > 0]
```

Out[•]=

$$\frac{e^{-2x^{-a}}\left(-1+e^{x^{-a}}\left(1-x^{-a}\right)\right)+\left(2-2\,e^{-x^{-a}}+x^{-2a}\right)\text{ExpIntegralE}\left[1,\,x^{-a}\right]}{2\,a\,x}+$$

$$\frac{e^{-x^{-a}}\left(-1+e^{-x^{-a}}+2\,\text{EulerGamma}-2\,\text{ExpIntegralEi}\left[-x^{-a}\right]-2\,a\,\text{Log}[x]\right)}{2\,a\,x}$$

In[•]:=
$$\text{FullSimplify}\left[\frac{e^{-2x^{-a}}\left(-1+e^{x^{-a}}\right)+\left(-1+e^{-x^{-a}}+x^{-a}\right)\text{ExpIntegralEi}\left[-x^{-a}\right]}{a\,x}+\frac{e^{-x^{-a}}\left(-1+e^{-x^{-a}}+\text{EulerGamma}-\text{ExpIntegralEi}\left[-x^{-a}\right]-a\,\text{Log}[x]\right)}{a\,x}\right]$$

Out[•]=
$$\frac{\left(-1+x^{-a}\right)\text{ExpIntegralEi}\left[-x^{-a}\right]+e^{-x^{-a}}\left(\text{EulerGamma}-a\,\text{Log}[x]\right)}{a\,x}$$

In[•]:=
$$\text{Integrate}\left[\frac{e^{-2x^{-a}}\left(-1+e^{x^{-a}}\left(1-x^{-a}\right)\right)+\left(2-2e^{-x^{-a}}+x^{-2a}\right)\text{ExpIntegralE}\left[1,x^{-a}\right]}{2\,a\,x}+\frac{e^{-x^{-a}}\left(-1+e^{-x^{-a}}+2\,\text{EulerGamma}-2\,\text{ExpIntegralEi}\left[-x^{-a}\right]-2\,a\,\text{Log}[x]\right)}{2\,a\,x},\ \{x,0,\text{Infinity}\},\ \text{Assumptions}\to a>0\right]$$

Out[•]=
$$\text{Integrate}\left[\frac{e^{-2x^{-a}}\left(-1+e^{x^{-a}}\left(1-x^{-a}\right)\right)+\left(2-2e^{-x^{-a}}+x^{-2a}\right)\text{ExpIntegralE}\left[1,x^{-a}\right]}{2\,a\,x}+\frac{e^{-x^{-a}}\left(-1+e^{-x^{-a}}+2\,\text{EulerGamma}-2\,\text{ExpIntegralEi}\left[-x^{-a}\right]-2\,a\,\text{Log}[x]\right)}{2\,a\,x},\ \{x,0,\infty\},\ \text{Assumptions}\to a>0\right]$$

In[•]:=
```
Integrate[(H[x, y, a, ρ] - G[x, a] × Gy[y, a, ρ]) * (D[f[y, a], y] / x),
  {y, 0, x}, Assumptions → a > 0 && x > 0] + G[x, a] * Integrate[
  (1 - Gy[y, a, ρ]) * (D[f[y, a], y] / x), {y, x, Infinity}, Assumptions → a > 0 && x > 0]
```

Out[•]=
$$\text{Integrate}\left[\frac{-e^{-x^{-a}-y^{-a}}\left(1+y^{-a}\rho[0]\right)+e^{-y^{-a}}\left(1+y^{-a}\rho\left[\left(\frac{y}{x}\right)^{a}\right]\right)}{x\,y},\ \{y,0,x\},\ \text{Assumptions}\to a>0\ \&\&\ x>0\right]-\frac{e^{-x^{-a}}\left(-\text{EulerGamma}+\text{ExpIntegralEi}\left[-x^{-a}\right]+a\,\text{Log}[x]+\rho[0]-e^{-x^{-a}}\rho[0]\right)}{a\,x}$$

Schmier

In[•]:=
```
Integrate[y^(-a) Exp[-y^(-a)] / y, {y, 0, Infinity}, Assumptions → a > 0]
```

Out[•]=
$$\frac{1}{a}$$

In[•]:=
```
Integrate[y^(-a) Exp[-y^(-a)] * (-a) * y^(-a - 1), {y, 0, Infinity}, Assumptions → a > 0]
```

Out[•]= -1

In[•]:=
```
Integrate[y^(-a) Exp[-y^(-a)] * D[y^(-a) Log[y], y], {y, 0, Infinity}, Assumptions → a > 0]
```

Out[•]= $\frac{2 - \text{EulerGamma}}{a}$

In[•]:=
```
Integrate[Exp[-y^(-a)] / y, {y, 0, Infinity}, Assumptions → a > 0]
```

Integrate: Integral of $\frac{e^{-y^{-a}}}{y}$ does not converge on {0, ∞}.

Out[•]= $\text{Integrate}\left[\frac{e^{-y^{-a}}}{y}, \{y, 0, \infty\}, \text{Assumptions} \to a > 0\right]$

In[•]:=
```
z =.
```

In[•]:=
```
Integrate[(Exp[-x^(-a) / z] * z^(-2) * (1 - z) * x^(-a) - Exp[-x^(-a)]) / x,
 {z, 0, 1}, Assumptions → a > 0 && x > 0]
```

Out[•]= $-x^{-1-a}\,\text{Gamma}[0, x^{-a}]$

In[•]:=
```
Integrate[-x^(-1-a) Gamma[0, x^(-a)], {x, 0, ∞}, Assumptions → a > 0]
```

Out[•]= $-\frac{1}{a}$

In[•]:=
```
Integrate[(1 - z) * Exp[-x^(-a) / z] * z^(-2) * x^(-a), {z, 0, 1}, Assumptions → a > 0 && x > 0]
```

Out[•]= $e^{-x^{-a}} - x^{-a}\,\text{Gamma}[0, x^{-a}]$

In[•]:=
```
Integrate[-x^(-a) Gamma[0, x^(-a)] / x, {x, 0, Infinity}, Assumptions → a > 0 && x > 0]
```

Out[•]= $-\frac{1}{a}$

In[•]:=
```
Integrate[Exp[-x^(-a) / z] * z^(-2) * x^(-a),
 {z, 0, 1 - p}, Assumptions → a > 0 && x > 0 && 0 < p < 1]
```

Out[•]= $e^{\frac{x^{-a}}{-1+p}}$

In[•]:=
```
Integrate[(1 - z) / p * Exp[-x^(-a) / z] * z^(-2) * x^(-a),
 {z, 1 - p, 1}, Assumptions → a > 0 && x > 0 && 0 < p < 1]
```

Out[•]= $\frac{-e^{-x^{-a}}\left(-1 + e^{\frac{p\,x^{-a}}{-1+p}}\right) + x^{-a}\left(-\text{Gamma}[0, x^{-a}] + \text{Gamma}\left[0, -\frac{x^{-a}}{-1+p}\right]\right)}{p}$

In[•]:=
```
FullSimplify[(-e^(-x^(-a)) (-1 + e^(p x^(-a)/(-1+p))) + x^(-a) (-Gamma[0, x^(-a)] + Gamma[0, -x^(-a)/(-1+p)])) / p + e^(x^(-a)/(-1+p))]
```

Out[•]= $\frac{e^{-x^{-a}}\left(1 + e^{\frac{p\,x^{-a}}{-1+p}}(-1 + p)\right) + x^{-a}\left(-\text{Gamma}[0, x^{-a}] + \text{Gamma}\left[0, -\frac{x^{-a}}{-1+p}\right]\right)}{p}$

In[•]:=
```
p =.
```

In[•]:=
```
Integrate[(e^(-x^-a) (1 + e^(p x^-a/(-1+p)) (-1 + p)) + x^-a (-Gamma[0, x^-a] + Gamma[0, -x^-a/(-1+p)]))/p - Exp[-x^(-a)])/x,
 {x, 0, ∞}, Assumptions → a > 0 && 0 < p < 1]
```

Out[•]= $\frac{-p+(-1+p)\,\mathrm{Log}[1-p]}{a\,p}$

In[•]:=
```
Integrate[x^(-a - 1) (Exp[-x^(-a)/z]), {x, 0, Infinity}, Assumptions → a > 0 && z > 0]
```

Out[•]= $\frac{z}{a}$

In[•]:=
```
Integrate[Gamma[0, x^-a]/x, {x, 0, Infinity}, Assumptions → a > 0]
```

Integrate: Integral of $\frac{\mathrm{Gamma}[0, x^{-a}]}{x}$ does not converge on $\{0, \infty\}$.

Out[•]= $\mathrm{Integrate}\left[\frac{\mathrm{Gamma}[0, x^{-a}]}{x}, \{x, 0, \infty\}, \mathrm{Assumptions} \to a > 0\right]$

In[•]:=
```
Integrate[(e^(-x^-a/z) - e^(-x^-a))/x, {x, 0, Infinity}, Assumptions → a > 0 && z > 0]
```

Out[•]= $\frac{\mathrm{Log}[z]}{a}$

In[•]:=
```
D[Exp[-x^(-a)/z] (1 + x^(-a)/z) Log[x^(-a)/z] + Gamma[0, x^(-a)/z] + Exp[-x^(-a)/z], z] //
 FullSimplify
```

Out[•]= $\frac{e^{-\frac{x^{-a}}{z}}\, x^{-2a}\, \mathrm{Log}\left[\frac{x^{-a}}{z}\right]}{z^3}$

In[•]:=
```
Integrate[Exp[-x^(-a)/z] (1 + x^(-a)/z) Log[x^(-a)/z] + Gamma[0, x^(-a)/z] + Exp[-x^(-a)/z],
 {z, 0, 1}, Assumptions → a > 0 && x > 0]
```

Out[•]= $(-1+x^{-a})\,\mathrm{ExpIntegralEi}[-x^{-a}] + e^{-x^{-a}}\,(1 - a\,\mathrm{Log}[x])$

In[•]:=
```
Integrate[(Exp[-x^(-a)/z] (1 + x^(-a)/z) Log[x^(-a)/z] + Gamma[0, x^(-a)/z] +
      Exp[-x^(-a)/z] - ((-1 + x^-a) ExpIntegralEi[-x^-a] + e^(-x^-a) (1 - a Log[x])))/x,
 {x, 0, Infinity}, Assumptions → a > 0 && z > 0]
```

Out[•]= $-\frac{(-1+\mathrm{EulerGamma})\,(1+\mathrm{Log}[z])}{a}$

In[•]:=
```
Integrate[(x^-a ExpIntegralEi[-x^-a])/x, {x, 0, Infinity}, Assumptions → a > 0]
```

Out[•]= $-\frac{1}{a}$

In[∘]:=
```
Integrate[(e^(-x^a) (1 - EulerGamma - a Log[x])) / x, {x, 0, Infinity}, Assumptions → a > 0]
```

Integrate: Integral of $-\frac{e^{-x^a}(-1+\text{EulerGamma}+a\,\text{Log}[x])}{x}$ does not converge on {0, ∞}.

In[∘]:=
```
Integrate[(ExpIntegralEi[-x^-a] - e^(-x^a) EulerGamma) / x,
 {x, 0, Infinity}, Assumptions → a > 0]
```

Integrate: Integral of $-\frac{e^{-x^a}\,\text{EulerGamma}}{x}+\frac{\text{ExpIntegralEi}[-x^{-a}]}{x}$ does not converge on {0, ∞}.

Out[∘]= $\text{Integrate}\left[\frac{-e^{-x^a}\,\text{EulerGamma}+\text{ExpIntegralEi}[-x^{-a}]}{x}, \{x, 0, \infty\}, \text{Assumptions} \to a > 0\right]$

```
Integrate[((-1 + x^-a) ExpIntegralEi[-x^-a] + e^(-x^a) (1 - a Log[x]) + Gamma[0, x^(-a)] +
    (2 - EulerGamma) Exp[-x^(-a)]) / x, {x, 0, Infinity}, Assumptions → a > 0]
```

In[∘]:=
```
Integrate[e^(-x^a) x^(-a) D[x^(-a) Log[x], x], {x, 0, Infinity}, Assumptions → a > 0]
```

Out[∘]= $\frac{2-\text{EulerGamma}}{a}$

In[∘]:=
```
Limit[Exp[-x^(-a) / Abs[z]] (1 + x^(-a) / Abs[z]) Log[x^(-a) / Abs[z]] +
   Gamma[0, x^(-a) / Abs[z]] + Exp[-x^(-a) / Abs[z]], z → 0, Assumptions → x > 0 && a > 0]
```

Out[∘]= 0

In[∘]:=
```
Integrate[(ExpIntegralEi[-x^(-a)] (x^(-a) - 1) + (EulerGamma - a Log[x]) Exp[-x^(-a)]) / x,
 {x, 0, Infinity}, Assumptions → a > 0]
```

Out[∘]= $\frac{-6+\pi^2}{6\,a}$

noch mehr schmier

In[∘]:=
```
Limit[Exp[-x / Abs[z]] (1 + x / Abs[z]), z → 1, Assumptions → x > 0]
```

Out[∘]= $e^{-x}(1+x)$

In[∘]:=
```
Integrate[Exp[-y^(-a) / z] y^(-a) / y, {y, 0, Infinity}, Assumptions → a > 0 && z > 0]
```

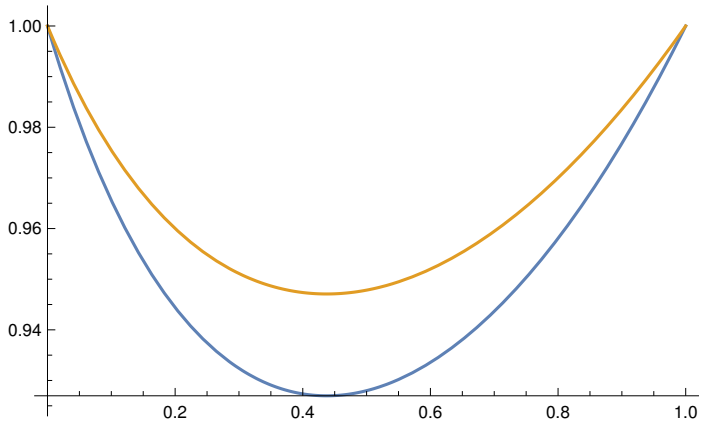
Out[∘]= $\frac{z}{a}$

In[∘]:=
```
Integrate[Exp[-y^(-a)] y^(-a) * D[y^(-a * w), y],
 {y, 0, Infinity}, Assumptions → a > 0 && 1 > w > 0]
```

Out[∘]= -w Gamma[1 + w]

In[•]:=
```
Integrate[Exp[-y^(-a)] y^(-a) * D[y^(-a * w) Log[y], y],
  {y, 0, Infinity}, Assumptions → a > 0 && 1 > w > 0]
```

Out[•]=
$$\frac{\text{Gamma}[1+w]\left(1+w\,\text{PolyGamma}[0,\,1+w]\right)}{a}$$

In[•]:=
```
Plot[{wp[ppp] Gamma[1 + wp[ppp]], wp[ppp]}, {ppp, 0, 1}]
```

Out[•]=